ABHIMANYU MANDAL

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PROFILE

Computational biologist with research experience in genomics, transcriptomics, whole-exome sequencing (WES), and multi-omics data analysis. Experienced in developing reproducible bioinformatics pipelines using R, Python, and Linux for the analysis of large-scale biological datasets. Interested in developing scalable bioinformatics workflows and AI-driven approaches for genomics, precision medicine, and biomarker discovery.

TECHNICAL SKILLS

Genomics & Variant Analysis: Whole-exome sequencing (WES), variant calling, and annotation, GATK, IGV, ClinVar, gnomAD, OMIM, ACMG/AMP variant classification

Transcriptomics: Bulk RNA-seq, single-cell RNA-seq (Seurat), Differential gene expression, clustering, PCA, UMAP, t-SNE, Pathway enrichment analysis

Programming: R (advanced), Python, Bash, SQL

Workflow & Infrastructure: Linux, Bash, Git, GitHub, Nextflow DSL2, Docker, HPC (SLURM)

Visualization: R Shiny, ggplot2, Tableau

RESEARCH EXPERIENCE

Project Research Assistant, NDMC, IIT Bombay *Sep 2024 – Mar 2025*

Computational epidemiology – Bayesian spatiotemporal modelling, R, HPC

- Built Bayesian spatiotemporal risk models across 70+ districts in Maharashtra and Gujarat, directly informing district-level TB risk stratification used by national public health stakeholders.
- Processed and analysed 10,000+ clinical and public health records in R, delivering fully reproducible pipeline outputs and structured technical reports under NDMC program timelines.
- Designed interactive dashboards and heatmaps that translated complex epidemiological findings for non-technical policy audiences.

Bioinformatics Research Intern, Hu Lab, University of Western Ontario *May 2023 – Aug 2023*

Cancer single-cell genomics – Seurat, scRNA-seq, differential expression, pathway analysis

- Developed a computational drug-repurposing workflow using **single-cell RNA-seq data from 28,441 cells**, integrating normal breast epithelial reference cells with naive and treated HER2+ breast cancer states.
- Performed **cell-type-specific differential expression, drug-response scoring, pathway enrichment, and comparative validation** to identify shared disease-state, treatment-specific, and transition-associated therapeutic candidates.
- **Reconstructed and extended the original 2023 analysis** in **2026** to improve reproducibility, validate the workflow, and evaluate therapeutic candidates using updated computational analyses.

Biomedical Data Analyst Intern, Elucidata Corporation *Feb 2022 – Jul 2022*

Multi-omics data infrastructure – R Shiny, QC pipelines, technical documentation

- Developed an R Shiny platform for quality assessment, visualization, and exploration of more than 1,000 multi-omics datasets, reducing manual QC time by approximately 30%.
- Performed systematic data quality checks and exploratory analysis, developing clear technical documentation that improved reproducibility across the data science team.

EDUCATION

Integrated Dual Degree (B.Tech + M.Tech), Pharmaceutical Engineering & Technology, IIT (BHU), Varanasi *2019 – 2024*

CGPA: 8.74 / 10 | Specialisation: Genomics & Computational Biology

- Master’s Thesis: Germline Variant Analysis in Indian OSCC-GB Patients (WES) — analysed 300+

WES samples, applied GATK/ClinVar/gnomAD annotation and ACMG/AMP classification to identify rare high-impact pathogenic variants linked to early onset and poor prognosis (manuscript under peer review).

SELECTED PROJECTS

Breast Cancer scRNA-seq Drug Repurposing - Expanded Reanalysis - GitHub

- **Reconstructed and extended a breast cancer scRNA-seq workflow** using Seurat, pseudobulk differential expression, disease-signature construction, and LINCS connectivity analysis.
- Screened **9,108 perturbational signatures across 5 breast cancer cell lines** and prioritized **66 computational candidates** using ChEMBL mechanism annotation and external literature evidence.

Nextflow NGS DSL2 Germline Variant Calling Pipeline - GitHub

- Developed a modular **Nextflow DSL2** pipeline implementing the **GATK Best Practices** workflow for germline variant calling, automating FASTQ preprocessing, BWA-MEM alignment, SAMtools processing, GATK variant calling, and Ensembl VEP annotation to deliver reproducible and scalable NGS analysis.

Nextflow NGS Quality Control Pipeline - GitHub

- Developed a modular Nextflow DSL2 workflow for automated NGS quality control using FastQC and MultiQC, implementing reusable processes and structured outputs to improve reproducibility and scalability.

Drug Neurotoxicity Prediction - GitHub

- Developed an end-to-end RDKit-based machine learning pipeline for neurotoxicity prediction, achieving **84.06% accuracy and 0.91 ROC-AUC.**

Cancer Genomics – WES Variant Analysis - GitHub

- Identified high-impact pathogenic variants in oncology WES datasets using multi-database annotation workflows; improved pipeline classification accuracy by integrating ClinVar, gnomAD, and multiple clinical prediction tools.

Healthcare Claims Analytics (R & Tableau) - GitHub

- Analyzed 15,000 health insurance claims using R and Tableau, building interactive dashboards for settlement, rejection, fraud, and hospital performance analysis, and generating actionable insights through EDA, KPI tracking, and business intelligence reporting.

PUBLICATIONS

- Mandal A., Khattri A. Predisposing Germline Mutations Associated with Early Onset and Poor Prognosis in Indian OSCC-GB Patients. (Under Review)
- Nag S., Mandal A., Joshi A., et al. Sialyltransferases and Neuraminidases: Potential Targets for Cancer Treatment. Diseases. 2022; 10(4):114.
- Nag S., Baidya ATK, Mandal A., et al. Deep learning tools for advancing drug discovery and development. 3 Biotech. 2022; 12(5).

AWARDS & FELLOWSHIPS

- Project Research Assistant Fellowship – National Disease Modelling Consortium, IIT Bombay (2024)
- Junior Research Fellowship – IIT Bombay (2024)
- Mitacs Globalink Research Internship – University of Western Ontario, Canada (2023)
- Biomedical Data Analyst Fellowship – Elucidata Corporation (2022)

LEADERSHIP & MENTORSHIP

- Mentored 20+ undergraduates on research methodology, project planning, and graduate applications at IIT-BHU Research Cell (2022–2024), building the teaching and supervision skills valued in a multi-PI lab environment.
- Led student teams to curate 1,000+ research opportunities and coordinated alumni outreach, developing the organisational and communication skills directly applicable to collaborative, multidisciplinary research settings.